MIL-F-3222E
26 October 1978
SUPERSEDING
MIL-F-3222D
20 April 1971

MILITARY SPECIFICATION

FLOODLIGHTS AND LANTERNS, PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for methods, materials, and procedures, for the preservation, packing, and marking of floodlights and lanterns (except trailer-mounted floodlights) (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

FSC PACK

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 101	- Preservation, Packaging, and Packing Materials: Test Procedures.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article.

3.1.1 Preproduction pack. Unless otherwise specified (see 6.2), the contractor shall furnish a preproduction pack for examination and testing within the time frame specified (see 6.2), to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be prepared utilizing either the preproduction model or a production floodlight or lantern. When the preproduction model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the preproduction model and production floodlights or lanterns.

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3.2 Material. Materials shall be as specified herein and in applicable specifications and standards. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects (see 6.6).

3.3 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts vulnerable to damage.

3.3.1.2 Methods. Methods of preservation indicated by symbols herein shall conform to the corresponding symbols in MIL-P-116.

3.3.1.3 Preservatives. Preservatives shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

3.3.1.4 Unprotected metal surfaces. Preservatives shall not be applied to reflectors. Exterior unprotected metal surfaces of the floodlights, lanterns, and accessories, requiring the application of a contact preservative in accordance with the preservative application criteria of MIL-P-116, shall be coated with Type P-19 preservative. Interior surfaces, machined surfaces, and threaded surfaces, if unprotected, and requiring the application of a contact preservative in accordance with the preservative application criteria of MIL-P-116, shall be coated with Type P-7 preservative. Excess preservative shall be allowed to drain. Exposed surfaces coated with Type P-7 preservative shall be wrapped or covered with greaseproof barrier material conforming to MIL-B-121, Type II, Grade A, Class 2, and the wrap or cover secured in place with tape conforming to PPP-T-60, Type IV.

3.3.1.5 Repair parts and accessories. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve the repair parts and accessories. When sufficient space is available, the preserved repair parts and accessories shall be placed within the container with the end item for which they are intended. When sufficient space is not available, for placing with the end item, the preserved repair parts and accessories shall be consolidated as specified in 3.3.1.11.

3.3.1.6 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, Level A. When sufficient space is available, the preserved tools shall be placed within the container containing the end item for which they are intended. When sufficient space is not available, for placing with the end item, the preserved tools shall be consolidated as specified in 3.3.1.11.

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3.3.1.7 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1. When space is available the preserved publications shall be placed within the container with the end item. When space is not available for placing with the end item, the preserved publications shall be consolidated as specified in 3.3.1.11.

3.3.1.8 Electric operated and battery operated floodlights and lanterns.

3.3.1.8.1 Enclosed, gasketed, and weatherproofed. Each complete floodlight and each complete lantern shall be preserved in accordance with MIL-P-116, Method III. Containers shall conform to PPP-B-636, W5c, W6c, or V3c, as applicable, Style RSC. Cushioning shall be provided to immobilize the items within the containers. Box closure shall be in accordance with the appendix to the box specification, Method V.

3.3.1.8.2 Not enclosed, not gasketed, and not weatherproofed. Each complete floodlight and each complete lantern shall be preserved in accordance with MIL-P-116, Method IA-14. When a rigid carrying case is furnished it may be used as the inner container. Cushioning shall be provided within the inner containers and in rigid carrying cases to immobilize the items within the containers. The outer containers shall conform to PPP-B-636, W5c, W6c, or V3c, as applicable, Style RSC. Box closure shall be in accordance with the appendix to the box specification, Method V.

3.3.1.9 Carbide, acetylene, and other gas-illuminated floodlights and lanterns. Each complete floodlight and each complete lantern shall be preserved in accordance with MIL-P-116, Method IC-2.

3.3.1.10 Gasoline and oil-burning floodlights and lanterns. Removed mantles and replacement mantles for each item shall be preserved in a kraft envelope and the envelope sealed. Each complete floodlight and each complete lantern, with preserved mantles, shall be preserved in accordance with MIL-P-116, Method IC-2.

3.3.1.11 Consolidation. Repair parts, tools, technical publications, and accessories such as mounting masts, tripod mountings, and globes for each floodlight or lantern, that is not preserved in the container with the end item, shall be consolidated in a close-fitting box conforming to PPP-B-636, W5c or W6c, Style RSC. Cushioning shall be provided to immobilize the items within the boxes. Box closure shall be in accordance with the appendix to the box specification, Method V.

3.3.2 Level B. Unless otherwise specified (see 6.5), Level B preservation shall be as follows: Preservatives shall not be applied to reflectors. Removed mantles and replacement mantles shall be preserved as specified for Level A. Repair parts, tools, technical publications, and accessories shall

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be preserved as specified for Level A. Consolidation shall be as specified for Level A. Floodlights and lanterns in rigid carrying cases shall not require preservation. Each floodlight and each lantern not in a rigid carrying case shall be preserved in a box specified herein. Cushioning shall be provided and applied in a manner to prevent movement within the boxes.

3.3.3 Commercial. Floodlights, lanterns, repair parts, tools, and accessories shall be preserved in accordance with MIL-STD-1188.

3.4 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

3.4.1 Level A. Complete floodlights and lanterns, of like description, preserved as specified in 3.3, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, Style I or J. Strapping shall conform to QQ-S-781, Class 1, Type I or IV size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

3.4.2 Level B. Packing shall be as specified for Level A except that boxes shall be PPP-B-601, Domestic Type and strapping shall be Finish A.

3.4.3 Commercial. Floodlights, lanterns, repair parts, tools, and accessories shall be packed in accordance with MIL-STD-1188.

3.5 Marking.

3.5.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

3.5.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined as specified in 4.4.4. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Unless otherwise specified (see 6.2), the preproduction pack for Level A and B shall be tested in accordance with 4.3.2.1 or 4.3.2.2 as applicable to the gross weight of the box. Failure of the applicable test shall be cause for rejection.

4.3.2.1 Packs not exceeding 200 pounds gross weight. The pack shall be subjected to the free-fall drop test in accordance with Test Method 5007 of FED. TEST METHOD STD. No. 101.

4.3.2.2 Packs exceeding 200 pounds gross weight. The pack shall be subjected to the pendulum-impact test in accordance with Test Method 5012 of FED. TEST METHOD STD. No. 101.

4.3.3 Basis for rejection. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing or any visual damage to the contents shall constitute failure of the tests.

4.4 Quality conformance inspection.

4.4.1 Inspection stages. Inspection shall be in two stages as follows:

(a) The first stage shall include inspection of materials, methods, containers, and marking within the shipping container.
(b) The second stage shall include inspection of materials and marking after closing the shipping container.

4.4.2 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit of product.

4.4.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects marked "X" for the applicable level of packaging as specified in Table I. Acceptable Quality Level (AQL) shall be 2.5 percent defective.

Table I. Classification of Defects

Defect	Level A	Level B
101. Materials, methods, or containers not as specified. Each incorrect material, method, or container shall constitute one defect (3.2, 3.3, 3.4).	X	X
102. Surfaces not coated with preservatives as specified (3.3.1.4).	X	-
103. Coated surfaces not wrapped or covered as specified (3.3.1.4).	X	-
104. Box closure not Method V as specified (3.3.1.8.1, 3.3.1.8.2, 3.3.1.11).	X	-
105. Replacement mantles and removed mantles not preserved as specified (3.3.1.10, 3.3.2).	X	X
106. Consolidation not as specified (3.3.1.11, 3.3.2).	X	X
107. Cushioning not provided as specified (3.3.1.11, 3.3.2).	X	X
108. Strapping not as specified (3.4.1, 3.4.2).	X	X
109. Marking illegible, missing, incomplete, incorrect (3.5).	X	X

5. PACKAGING

(This section not applicable to this specification.)

6. NOTES

6.1 Intended use. This specification is intended to be used for reference in Section 5 of commodity specifications for floodlights and lanterns (except trailer-mounted floodlights) or for direct reference in contracts, and as an aid in the preparation of packaging data sheets.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction pack (see 3.1.1.).
(c) When a preproduction pack is not required (see 3.1.1).

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(d) Degree of preservation and degree of packing required (see 3.3 and 3.4). Level B preservation (see 3.3.2 and 6.5), is intended to provide limited but economical protection and should be specified only when it is determined the floodlights and lanterns will be held in covered storage.
(e) When Level B preservation is other than specified (see 3.3.2 and 6.5).
(f) When Finish A strapping is required (see 3.4.1).
(g) When preproduction pack testing is not required (see 4.3.2).

6.3 Trailer-mounted units. Preservation, packing, and marking of trailer-mounted floodlights will be in accordance with the packaging requirements of the applicable end item specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack shall be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the floodlights and lanterns in accordance with this specification.

6.5 Level B preservation. Level B preservation (see 3.3.2), has been developed for use by the Army and Air Force for the specific conditions specified in 6.2. For Navy procurements, details of Level B preservation, when required, should be developed at the time of procurement when all conditions are known.

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
Army - SM

User activity:
Navy - MC

Preparing activity:
Army - ME

Project PACK-0572

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-F-3222E Floodlights and Lanterns, Packaging of

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DATE

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